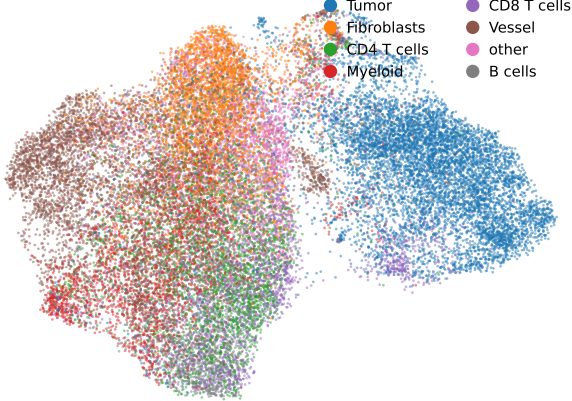


Figure 5 | Unsupervised token-space phenotype recovery: seed-unstable partition, robust vessel axis, no proliferation enrichment

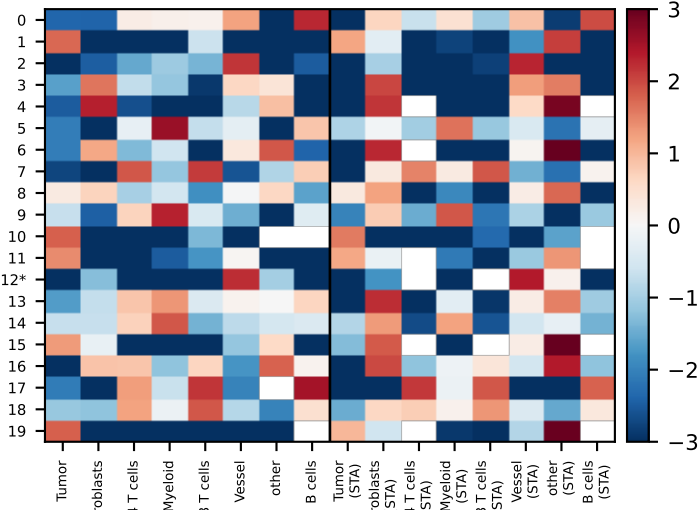
A1 Token clusters (UOP discovery, K=20)



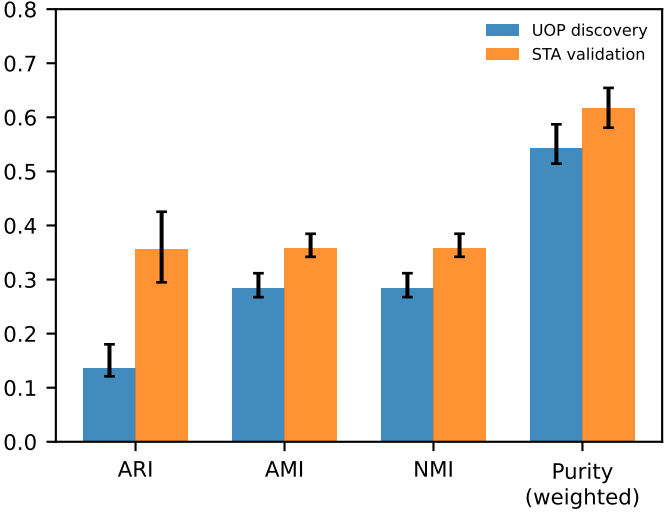
A2 Published cell labels (same cells)



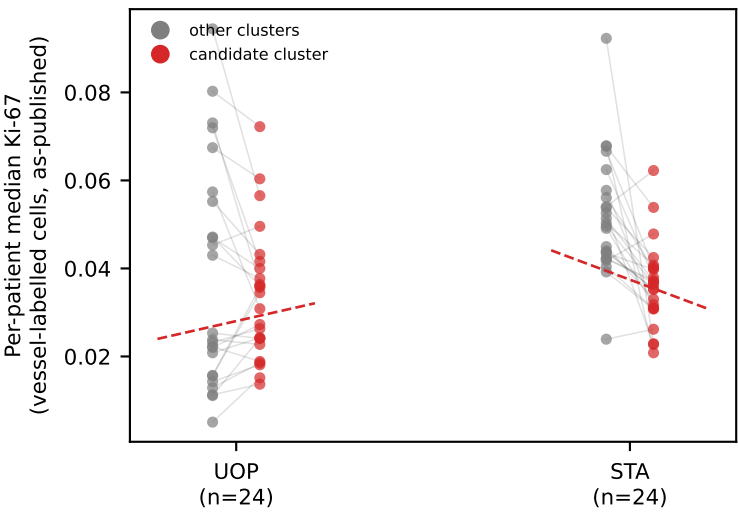
B Cluster × cell-type enrichment
log2 obs/exp (*=candidate; left UOP, right STA)



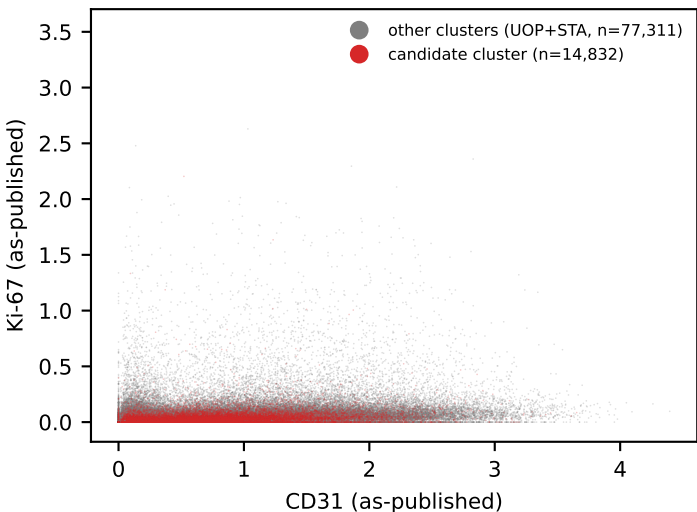
C External recovery of published labels
(patient-level bootstrap 95% CI, B=500)



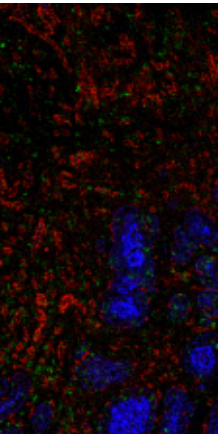
D Ki-67: candidate-cluster vessel cells vs
other-cluster vessel cells (paired patients)



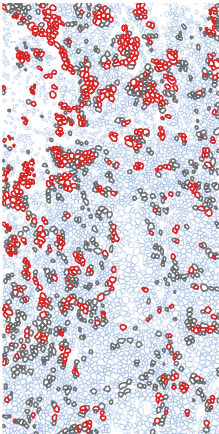
E Single-cell CD31-Ki-67 joint distribution
(vessel-labelled cells; grey=other, red=candidate)



F1 ROI OC12_003
R=CD31 G=Ki-67 B=panCK (raw, 0.99-norm)



F2 Cell outlines
red=candidate vessel; grey=other vessel; blue=others



Run: task3-v2-20260901
Clustering: MiniBatchKMeans K=20, UOP only
(273,408 cells, full fit), frozen VirTues tokens only
STA: nearest-centroid assignment (224,830)
Multi-seed stability: ARI 0.38–0.43 vs primary
(median 0.407; K=20 partition seed-unstable)
Vessel axis robust: vessel-dominated clusters
present in every seed and both cohorts
Candidate cluster 12 (criteria frozen in
dated reanalysis spec:
vessel%≥0.50 ∧ CD31 top-5 ∧ ≥22/24 patients)
Ki-67 paired (candidate – other vessel cells):
UOP –0.001 (p=0.41); STA –0.012 (p=8e-6)
→ no proliferation enrichment (direction negative)
Markers as-published (Einhaus values); bootstrap unit
= patient; UOP and STA analysed separately
Lymphatic identity not evaluable (no PROX1/LYVE1 in panel)